

P1) Look up the datasheet for the Analog AD8531 amplifier, find the component's specified values that would fully define the dominant pole (single pole) model introduced in the class, and complete the model. Note that the "Large Signal Voltage Gain" is effectively "open-loop gain" for our purposes. Using your model, estimate the open loop gain and phase of the amplifier at 100 Hz and 100 kHz. Hint: the open loop bandwidth is commonly not provided for amplifiers.

[https://www.analog.com/media/en/technical-documentation/data-sheets/AD8531\\_8532\\_8534.pdf](https://www.analog.com/media/en/technical-documentation/data-sheets/AD8531_8532_8534.pdf)

Note: assuming  $V_S=5V$ , so using table 2, figure 15, etc

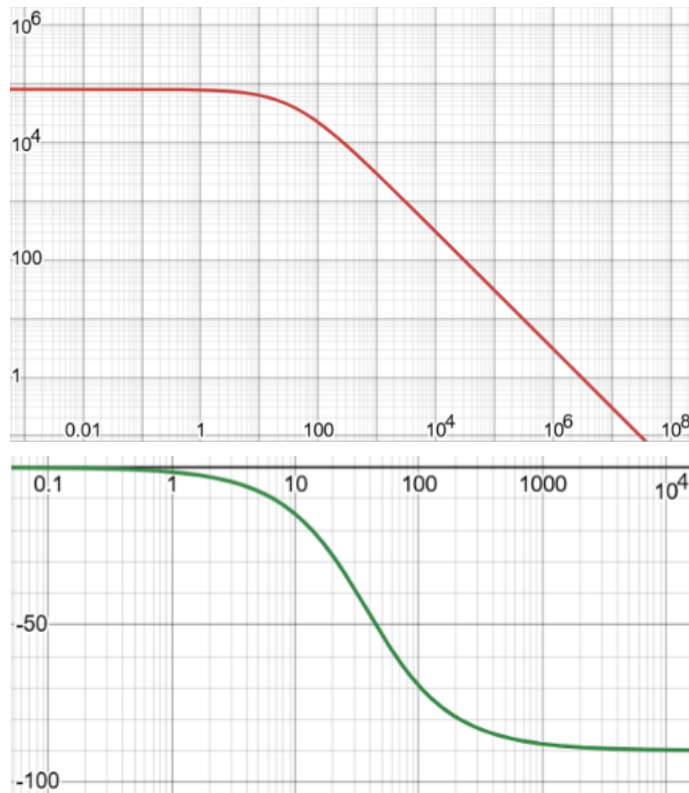
Large Signal Voltage Gain  $A_{VO} = 80V/mV = 80k = A_{ol}$

$GBP = f_t = 3MHz$

$f_B = f_t/A_0 = 3MHz/80k = 37.5Hz$

$A(jf) = A_{ol}/(1+jf/f_B) = 80k/(1+jf/37.5)$

$\phi(f) = -\arctan(f/f_B) = -\arctan(f/37.5)$



$A(100) = 80k/(1+(100/37.5)) = 21,818 \text{ V/V} = 86.78dB$

$\phi(100) = -\arctan(100/37.5) = -69.44deg$

$A(100k) = 80k/(1+(100k/37.5)) = 29.988 \text{ V/V} = 29.54dB$

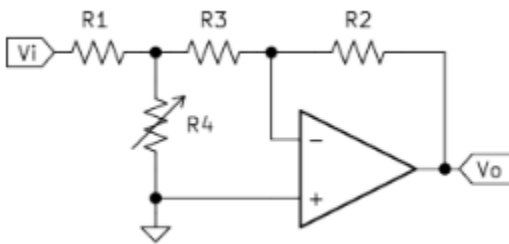
$\phi(100k) = -\arctan(100k/37.5) = -89.98deg$

P2) Calculate the frequency dependent closed-loop gain,  $A(jf)$ , for the Exam 1 Question 1 circuit (below);

assuming  $r_d = \infty$ ,  $r_o = 0$ , the resistors are valued as  $R_1 = R_3 = 5 \text{ k}\Omega$ ,  $R_2 = 1 \text{ M}\Omega$  and  $R_4$  is a  $10 \text{ k}\Omega$  potentiometer (wiper connected to ground). The op-amp used can be described with the model provided below.

(a) With  $R_4$  set to  $2.5 \text{ k}\Omega$ , calculate the frequency dependent closed-loop gain for this circuit, the operational bandwidth and closed-loop transition frequency. Sketch the open-loop and closed-loop gains vs. frequency and mark important points.

(b) The resistors are set such that when  $R_4 = 10 \text{ k}\Omega$ , a very large gain is achieved. What is the bandwidth at this gain?



$$a(jf) = \frac{500}{1 + jf/20 \text{ Hz}} \text{ V/mV}$$

a)

$$V_{out} = a(jf) (0 - V_n)$$

$$\text{if } V_i = 0: V_n = V_{out} (R_3 + (R_1 || R_4)) / (R_2 + R_3 + (R_1 || R_4))$$

$$\beta = V_{feedback} / V_{out} = V_n / V_{out} = (R_3 + (R_1 || R_4)) / (R_2 + R_3 + (R_1 || R_4))$$

$$\beta = (5\text{k} + (5\text{k} || 2.5\text{k})) / (1\text{M} + 5\text{k} + (5\text{k} || 2.5\text{k})) = 1/151$$

$$f_B = (1 + a_o \beta) f_y = (1 + 500\text{k}/151) * 20 = 66.24\text{kHz}$$

$$\text{if } V_{out} = 0: V_x = V_{in} (R_4 || (R_3 + R_2)) / ((R_4 || (R_3 + R_2)) + R_1) = 0.3327 V_{in}$$

$$V_n = V_x R_2 / (R_2 + R_3) = 0.3327 V_{in} * 1\text{M} / (1\text{M} + 5\text{k}) = 0.331126 V_{in} = V_{in} / 3.02$$

$$V_n = V_{in} / 3.02 + V_{out} / 151$$

$$V_{out} = a(jf) (0 - V_n)$$

$$V_n = -V_{out} / a(jf)$$

$$-V_{out} / a(jf) = V_{in} / 3.02 + V_{out} / 151$$

$$151 V_{out} + a(jf) V_{out} = -151 a(jf) V_{in} / 3.02$$

$$V_{out} = -151 a(jf) V_{in} / (3.02 (151 + a(jf)))$$

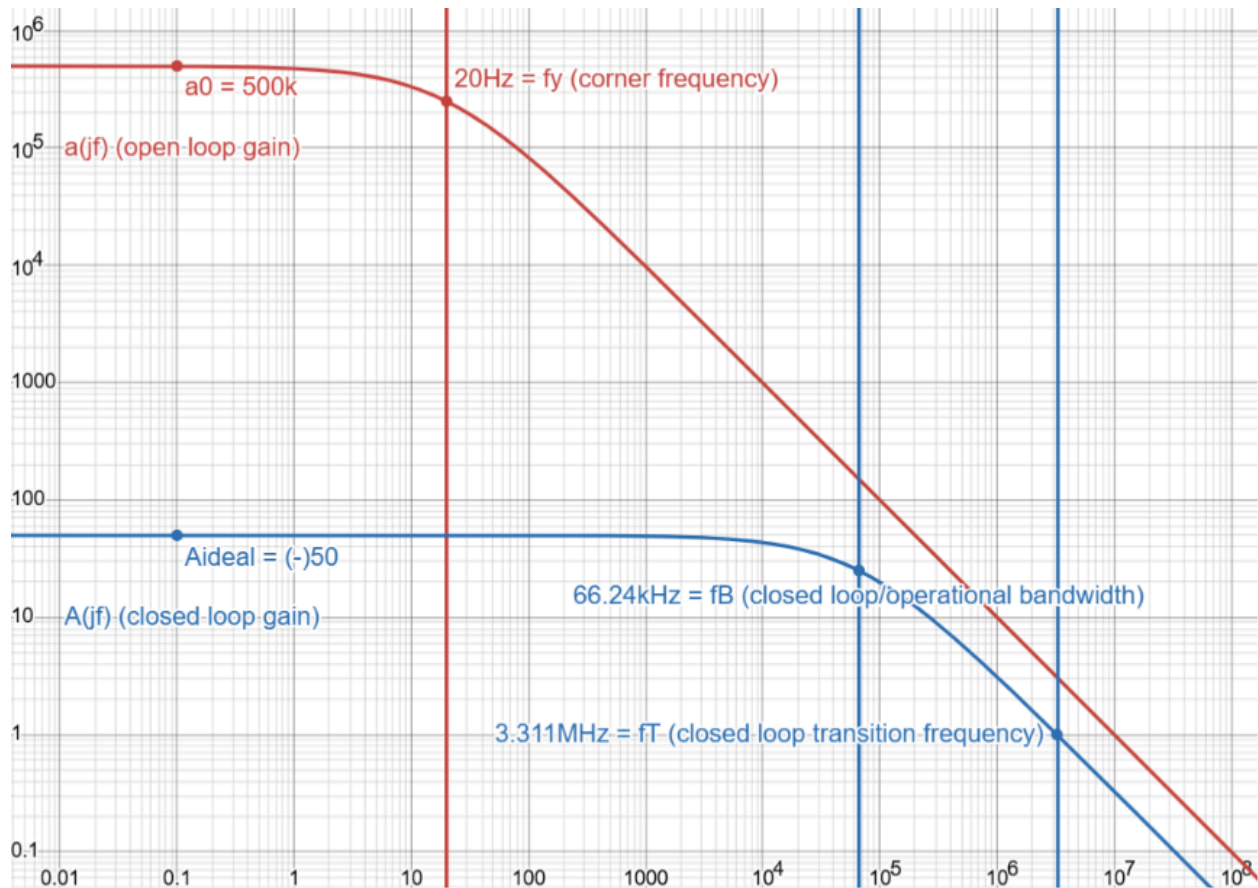
$$A = V_{out} / V_{in} = -151 a(jf) / (3.02 (151 + a(jf)))$$

$$A_{ideal} = -151 a(0) / (3.02 (151 + a(0))) = -151 * 500\text{k} / (3.02 (151 + 500\text{k})) = -49.9849$$

$$A_0 \approx A_{ideal} \text{ for low } f$$

$$f_t = |A_0| f_B = 50 * 66.24\text{kHz} = 3.311\text{MHz}$$

$$A(jf) = A_{ideal} / (1 + (jf/f_B)) = -50 / (1 + (jf/66.24\text{k}))$$



frequency dependent closed-loop gain:  $A(jf) = -50 / (1 + (jf/66.24\text{kHz}))$

operational bandwidth:  $f_B = 66.24\text{kHz}$

closed-loop transition frequency:  $f_T = 3.311\text{MHz}$

b)

$V_{out} = a(jf) (0 - V_n)$

$\beta = (5k + (5k || 10k)) / (1M + 5k + (5k || 10k)) = 1/121$

$f_B = (1 + a_0 \beta) f_y = (1 + 500k/121) \cdot 20 = 82.664\text{kHz}$

P3) The same op-amp as used in the question above is used to create a non-inverting amplifier circuit with a DC gain of 100. (a) What is the bandwidth of the circuit? Assemble a frequency-dependent single-pole model of the gain. (b) If the input resistance of the circuit at DC is  $R_i = 100 \text{ M}\Omega$ , determine  $R_i$  as  $f \rightarrow \infty$  and sketch  $R_i$  vs. frequency and mark important points.

a)

$$V_n = V_{in}$$

$$\beta = V_{feedback}/V_{out} = V_n/V_{out} = V_{in}/V_{out} = 1/A_{ideal} = 1/100$$

$$f_B = (1 + a_o \beta) f_y = (1 + 500k/100) 20\text{Hz} = 100\text{kHz}$$

$$A(jf) = A_{ideal}/(1 + jf/f_B) = 100/(1 + jf/100\text{kHz})$$

b)

if gain set to 0 ( $V_{in} = \text{test}$ ,  $V_{op} = 0$ ,  $V_{out} = 0$ ):

$$R_D = r_d + (R_1 \parallel (R_F + r_o))$$

if input shorted ( $V_{in} = 0$ ,  $V_{op} = \text{test}$ ,  $V_{out} = \text{NC}$ ):

$$V_N = V_{test} * (r_d \parallel R_1) / ((r_d \parallel R_1) + R_F + r_o), V_P = 0$$

$$TSC = a(jf) * (r_d \parallel R_1) / ((r_d \parallel R_1) + R_F + r_o)$$

if input open ( $V_{in} = \text{NC}$ ,  $V_{op} = \text{test}$ ,  $V_{out} = 0$ ):

$$V_{in} = V_P = V_N,$$

$$TOC = 0$$

$$R_i = R_D * (1 + TSC) / (1 + TOC)$$

$$R_i = (r_d + (R_1 \parallel (R_F + r_o))) * (1 + (a(jf) * (r_d \parallel R_1) / ((r_d \parallel R_1) + R_F + r_o)))$$

assume  $r_d \gg R_1, R_F$  and  $r_o \ll R_1, R_F$

$$R_i = r_d * (1 + (a(jf) * R_1 / (R_1 + R_F)))$$

$$A_{ideal} = 100 = 1 + (R_F/R_1) = (R_1 + R_F)/R_1$$

$$R_i = r_d * (1 + (a(jf)/A_{ideal}))$$

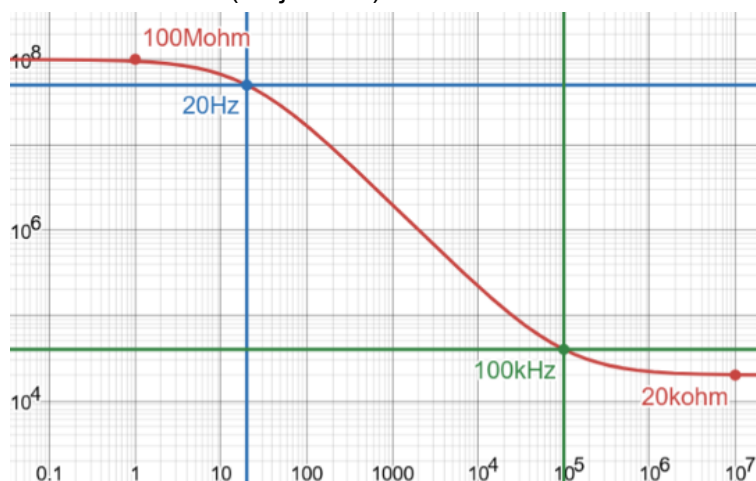
$$100\text{M} = r_d * (1 + (500k/100)) = r_d * 5k$$

$$r_d = 20k$$

$$R_i = 20k * (1 + (a(jf)/100))$$

$$R_i = 20k * (1 + (500k/(1 + jf/20\text{Hz}))/100))$$

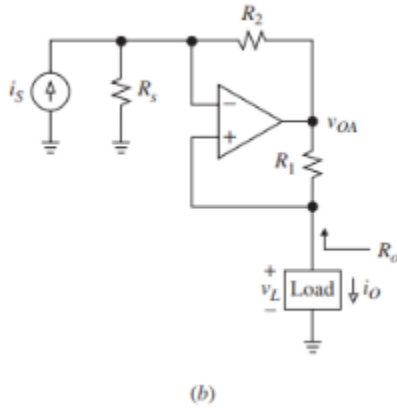
$$R_i = 20k + 100\text{M}/(1 + jf/20\text{Hz})$$



$R_i$  as  $f \rightarrow \infty$ :  $R_i \rightarrow 20\text{kohm}$

P4) Book Problem 6.25. Use exact analysis.

The grounded-load current amplifier of Fig. 2.12b is implemented with  $R_1 = R_2 = 10\text{ k}$  and an op amp having  $f_t = 10\text{ MHz}$ ,  $r_d = \infty$ , and  $r_o = 0$ . If the amplifier is driven by a source with a parallel resistance of  $30\text{ k}$  and drives a load of  $2\text{ k}$ , sketch and label the magnitude plots of the gain, the impedance seen by the source, and the impedance seen by the load.



a) gain

$$0 = I_S - I(R_S) - I(R_2)$$

$$I_S = I(R_S) + I(R_2)$$

$$I_S = V_n/R_S + (V_n - V_{op})/R_2$$

$$I_S = V_n/R_S + V_n/R_2 - V_{op}/R_2$$

$$I_S \cdot R_S \cdot R_2 = V_n \cdot R_2 + V_n \cdot R_S - V_{op} \cdot R_S$$

$$I_S \cdot R_S \cdot R_2 + V_{op} \cdot R_S = V_n \cdot R_2 + V_n \cdot R_S$$

$$V_n \cdot R_2 + V_n \cdot R_S = I_S \cdot R_S \cdot R_2 + V_{op} \cdot R_S$$

$$V_n \cdot (R_2 + R_S) = I_S \cdot R_S \cdot R_2 + V_{op} \cdot R_S$$

$$V_n = I_S \cdot R_S \cdot R_2 / (R_2 + R_S) + V_{op} \cdot R_S / (R_2 + R_S)$$

$$V_p = R_L \cdot I_{out}$$

$$V_{op} = a(jf) (V_p - V_n)$$

$$I_{out} = (V_{op} - V_p) / R_1$$

$$I_{out} \cdot R_1 = V_{op} - V_p$$

$$V_{op} = I_{out} \cdot R_1 + V_p$$

$$V_{op} = a(jf) (V_p - V_n)$$

$$V_{op} = a(jf) (V_p - (I_S \cdot R_S \cdot R_2 / (R_2 + R_S) + V_{op} \cdot R_S / (R_2 + R_S)))$$

$$I_{out} \cdot R_1 + V_p = a(jf) (V_p - (I_S \cdot R_S \cdot R_2 / (R_2 + R_S) + (I_{out} \cdot R_1 + V_p) \cdot R_S / (R_2 + R_S)))$$

$$I_{out} \cdot R_1 + R_L \cdot I_{out} = a(jf) (R_L \cdot I_{out} - (I_S \cdot R_S \cdot R_2 / (R_2 + R_S) + (I_{out} \cdot R_1 + R_L \cdot I_{out}) \cdot R_S / (R_2 + R_S)))$$

$$I_{out} \cdot R_1 + R_L \cdot I_{out} = a(jf) (R_L \cdot I_{out} - (I_S \cdot R_S \cdot R_2 / (R_2 + R_S) + (I_{out} \cdot R_1 + R_L \cdot I_{out}) \cdot R_S / (R_2 + R_S)))$$

$$I_{out} \cdot (R_1 + R_L) = a(jf) (R_L \cdot I_{out} - (I_S \cdot R_S \cdot R_2 / (R_2 + R_S) + (I_{out} \cdot R_1 + R_L \cdot I_{out}) \cdot R_S / (R_2 + R_S)))$$

$$I_{out} \cdot (R_1 + R_L) = a(jf) \cdot R_L \cdot I_{out} - (a(jf) \cdot I_S \cdot R_S \cdot R_2 / (R_2 + R_S) + a(jf) \cdot (I_{out} \cdot R_1 + R_L \cdot I_{out}) \cdot R_S / (R_2 + R_S))$$

$$I_{out} \cdot (R_1 + R_L) = a(jf) \cdot R_L \cdot I_{out} - (a(jf) \cdot I_S \cdot R_S \cdot R_2 / (R_2 + R_S) + a(jf) \cdot I_{out} \cdot (R_1 + R_L) \cdot R_S / (R_2 + R_S))$$

$$I_{out} \cdot (R_1 + R_L) = a(jf) \cdot R_L \cdot I_{out} - a(jf) \cdot I_S \cdot R_S \cdot R_2 / (R_2 + R_S) - a(jf) \cdot I_{out} \cdot (R_1 + R_L) \cdot R_S / (R_2 + R_S)$$

$$I_{out} \cdot (R_1 + R_L) - a(jf) \cdot R_L \cdot I_{out} = -a(jf) \cdot I_S \cdot R_S \cdot R_2 / (R_2 + R_S) - a(jf) \cdot I_{out} \cdot (R_1 + R_L) \cdot R_S / (R_2 + R_S)$$

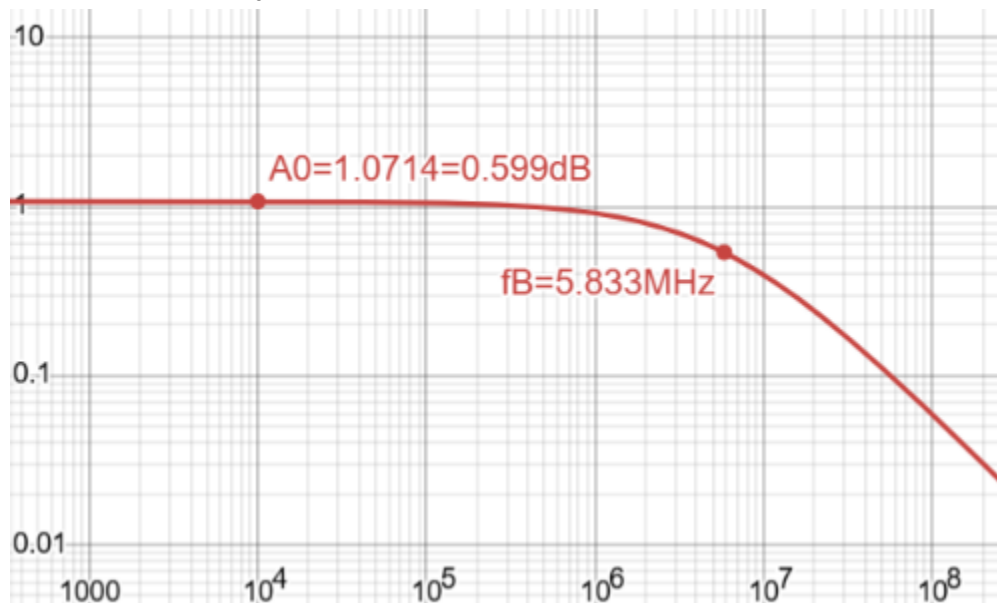
$$I_{out} \cdot (R_1 + R_L) - a(jf) \cdot R_L \cdot I_{out} + a(jf) \cdot I_{out} \cdot (R_1 + R_L) \cdot R_S / (R_2 + R_S) = -a(jf) \cdot I_S \cdot R_S \cdot R_2 / (R_2 + R_S)$$

$$\begin{aligned}
I_{out} * ( (R1+RL) - a(jf)*RL + a(jf)*(R1+RL)*RS/(R2+RS) ) &= - a(jf)*IS*RS*R2/(R2+RS) \\
I_{out} &= - a(jf)*IS*RS*R2 / ( (R2+RS)*( (R1+RL) - a(jf)*RL + a(jf)*(R1+RL)*RS/(R2+RS) ) ) \\
I_{out}/IS &= - a(jf)*RS*R2 / ( (R2+RS)*( (R1+RL) - a(jf)*RL + a(jf)*(R1+RL)*RS/(R2+RS) ) ) \\
I_{out}/IS &= - a(jf)*RS*R2 / ( (R2+RS)(R1+RL) - a(jf)*RL*(R2+RS) + a(jf)*(R1+RL)*RS ) \\
I_{out}/IS &= - RS*R2 / ( (R2+RS)(R1+RL)/a(jf) - RL*(R2+RS) + (R1+RL)*RS )
\end{aligned}$$

$$\begin{aligned}
I_{out}/IS &= - 30k*10k / ( (10k+30k)(10k+2k)/a(jf) - 2k*(10k+30k) + (10k+2k)*30k ) \\
I_{out}/IS &= - 30k*10k / ( 40k*12k/a(jf) - 2k*40k + 12k*30k ) \\
I_{out}/IS &= - 30*10 / ( 40*12/a(jf) - 2*40 + 12*30 ) \\
I_{out}/IS &= - 30 / ( 4*12/a(jf) - 2*4 + 12*3 ) \\
I_{out}/IS &= - 30 / ( 48/a(jf) - 8 + 36 ) \\
I_{out}/IS &= - 30 / ( 48/a(jf) + 28 )
\end{aligned}$$

$$\begin{aligned}
a(jf) &= a0/(1 + jf/fb) \\
I_{out}/IS &= - 30 / ( 48*(1 + jf/fb)/a0 + 28 ) \\
A &= - 30 / ( 48*(1 + jf/fb)/a0 + 28 ) \\
A &= - 30 / ( 48/a0 + 48*jf/(fb*a0) + 28 ) \\
A &= - 30 / ( 48*jf/(fb*a0) + 28 ) \\
A &= - 30 / ( 28 + 48*jf/(fb*a0) ) \\
A &= - 1.07142857143 / ( 1 + 1.71428571429*jf/(fb*a0) ) \\
A &= - 1.0714 / ( 1 + 1.714*jf/(fb*a0) ) \\
A &= - 1.0714 / ( 1 + jf/(fb*a0/1.714) ) \\
fB &= fb*a0/1.714 = ft/1.714 = 10MHz/1.714 = 5.833MHz
\end{aligned}$$

$$A = - 1.0714 / ( 1 + jf/5.833MHz )$$



b) source impedance

if gain set to 0 ( $V_{in}=test$ ,  $V_{op}=0$ ,  $V_{out}=0$ ,  $R_S$  ignored):

$R_D = R_2 = 10k$

if input shorted ( $V_{in}=0$ ,  $V_{op}=test$ ,  $V_{out}=0$ ):

$V_p = V_{test} R_L/(R_L+R_1)$ ,  $V_n=0$

$V_{op} = (V_p-V_n)a(jf) = V_{test} a(jf) R_L/(R_L+R_1)$

$TSC = -V_{op}/V_{test} = -a(jf) R_L/(R_L+R_1) = -0.1666 a(jf)$

if input open ( $V_{in}=NC$ ,  $V_{op}=test$ ,  $V_{out}=0$ ):

$V_p = V_{test} R_L/(R_L+R_1)$ ,  $V_n=V_{test}$

$V_{op} = (V_p-V_n)a(jf) = V_{test} a(jf) R_L/(R_L+R_1) - V_{test} a(jf) = V_{test} a(jf) (R_L/(R_L+R_1) - 1)$

$TOC = -V_{op}/V_{test} = -a(jf) (R_L/(R_L+R_1) - 1) = 0.8333 a(jf)$

$R_i = R_D(1+TSC)/(1+TOC)$

$R_i = 10k(1 + -0.1666 a(jf))/(1 + 0.8333 a(jf))$

$R_i = 10k(1 + -0.1666a_0/(1+jf/f_b))/(1 + 0.8333 a_0/(1+jf/f_b))$

$R_i = 10k((1+jf/f_b) + -0.1666a_0)/((1+jf/f_b) + 0.8333 a_0)$

$R_i = 10k(-0.1666/0.8333)(1/(-0.1666)+jf/(-0.1666 f_b) + a_0)/(1/(0.8333)+jf/(0.8333 f_b) + a_0)$

$R_i = -2k(1/(-0.1666)+jf/(-0.1666 f_b) + a_0)/(1/(0.8333)+jf/(0.8333 f_b) + a_0)$

$R_i = -2k(1/(-0.1666 a_0)+jf/(-0.1666 a_0 f_b) + 1)/(1/(0.8333 a_0)+jf/(0.8333 a_0 f_b) + 1)$

$R_i = -2k(jf/(-0.1666 a_0 f_b) + 1)/(jf/(0.8333 a_0 f_b) + 1)$

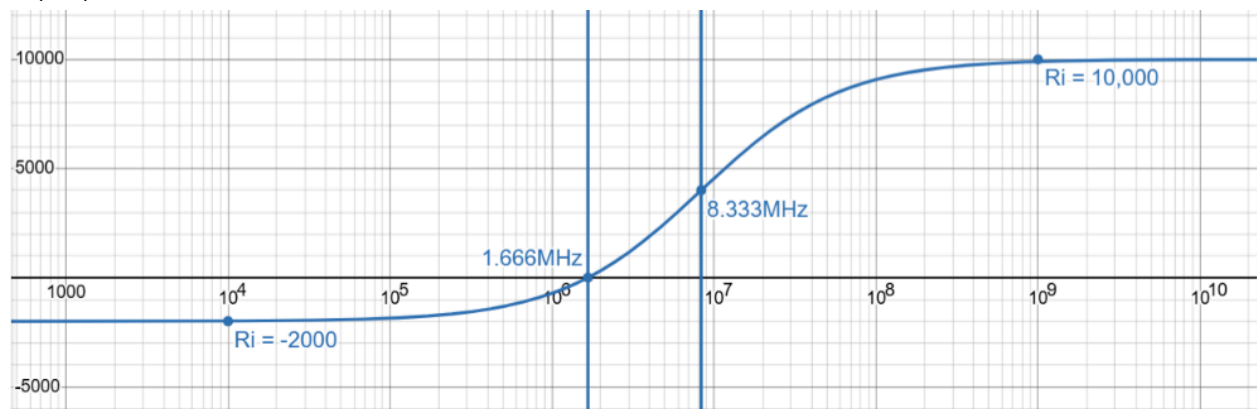
$R_i = -2k(1 - jf/(0.1666 a_0 f_b))/(1 + jf/(0.8333 a_0 f_b))$

$R_i = -2k(1 - jf/(0.1666 ft))/(1 + jf/(0.8333 ft))$

$R_i = -2k(1 - jf/1.666MHz)/(1 + jf/8.333MHz)$

$R_i(0) = -2k$

$R_i(\infty) = -10k$



c) load impedance

if gain set to 0 ( $V_{in}=NC$ ,  $V_{op}=0$ ,  $V_{out}=V_{test}$ ):

$$R_D = R_1 = 10k$$

if output shorted ( $V_{in}=NC$ ,  $V_{op}=test$ ,  $V_{out}=0$ ):

$$V_p = 0, V_n = V_{test} R_S / (R_S + R_2)$$

$$V_{op} = (V_p - V_n) a(jf) = V_{test} (-a(jf) R_S / (R_S + R_2))$$

$$TSC = -V_{op} / V_{test} = a(jf) R_S / (R_S + R_2) = 0.75 a(jf)$$

if output open ( $V_{in}=NC$ ,  $V_{op}=test$ ,  $V_{out}=NC$ ):

$$V_p = V_{test}, V_n = V_{test} R_S / (R_S + R_2)$$

$$V_{op} = (V_p - V_n) a(jf) = V_{test} a(jf) (1 - R_S / (R_S + R_2))$$

$$TOC = -V_{op} / V_{test} = -a(jf) (1 - R_S / (R_S + R_2)) = -0.25 a(jf)$$

$$R_o = R_D (1 + TSC) / (1 + TOC)$$

$$R_{io} = 10k (1 + 0.75 a(jf)) / (1 - 0.25 a(jf))$$

$$R_o = 10k (1 + 0.75 a_0 / (1 + jf/f_b)) / (1 - 0.25 a_0 / (1 + jf/f_b))$$

$$R_o = 10k ((1 + jf/f_b) + 0.75 a_0) / ((1 + jf/f_b) - 0.25 a_0)$$

$$R_o = 10k (0.75 / (-0.25)) (1 / (0.75) + jf / (0.75 f_b) + a_0) / (1 / (-0.25) + jf / (-0.25 f_b) + a_0)$$

$$R_o = -30k (1 / (0.75) + jf / (0.75 f_b) + a_0) / (1 / (-0.25) + jf / (-0.25 f_b) + a_0)$$

$$R_o = -30k (1 / (0.75 a_0) + jf / (0.75 a_0 f_b) + 1) / (1 / (-0.25 a_0) + jf / (-0.25 a_0 f_b) + 1)$$

$$R_o = -30k (jf / (0.75 a_0 f_b) + 1) / (jf / (-0.25 a_0 f_b) + 1)$$

$$R_o = -30k (1 + jf / (0.75 a_0 f_b)) / (1 - jf / (0.25 a_0 f_b))$$

$$R_o = -30k (1 + jf / (0.75 f_t)) / (1 - jf / (0.25 f_t))$$

$$R_o = -30k (1 + jf / (7.5MHz)) / (1 - jf / (2.5MHz))$$

$$R_o(0) = -30k$$

$$R_o(\infty) = 10k$$

